

Manish Krishna Kandrakota

Seeking Summer 2027 Software Engineering & Security Internships • Pittsburgh, PA
manish07070707@gmail.com • (412) 606-7948 • linkedin.com/in/manish-k-kandrakota • github.com/MANISH-K-07

EDUCATION

Carnegie Mellon University

Master of Science in Information Security

Pittsburgh, PA

August 2026 - Present

Sreenidhi Institute of Science and Technology

Bachelor of Technology in Computer Science and Engineering

Hyderabad, India

October 2022 - April 2026

- GPA: 8.32/10.0

EXPERIENCE

Honda Research Initiative

Independent Study (14-708, 12 units); Co-X Knowledge Management AI System

Pittsburgh, PA

August 2026 - Present

- Shortlisted for a highly competitive project with Honda Research (PI Prof. David Lamont), crafting a 3-state confirmation model (Confirmed/Proposed/Open) for the Tepper School of Business MS Product Management Capstone

SphinxATS: Multi-Attribute AI-Based Decision Support System

IEEE ICSSAS 2026

Erode, India

May 2026

- Lead author; modeled resume screening as a tunable-weight (60/30/10) multi-criteria decision problem - semantic similarity, skill overlap, and experience fit, with an attribute-level explainability layer for auditable ranking justifications
- Achieved 96.0% matching accuracy and a 0.960 F1 score across a curated 150-pair synthetic benchmark, a statistically significant gain ($p < 0.0001$) over keyword-based (93.3%) and TF-IDF-only (89.3%) baselines

Checkstyle: Java Static Analysis Framework

Open-Source Contributor (100+ merged pull requests)

Remote

December 2023 - May 2024

- Delivered rule enhancements, architectural refactoring, and reliability-critical bug fixes on a mature codebase used in enterprise CI pipelines, collaborating with 5 senior maintainers on design review and CI modernization

Charity with Clarity: Crowdfunding Using Smart Contracts

IEEE ICRISST 2024

Bengaluru, India

March 2024

- Spearheaded implementation, designing core Solidity contract logic (sendEth, createRequest, voteRequest, makePayment) enforcing majority-vote fund authorization and duplicate-payment prevention, validated via 9 Remix IDE test cases

PROJECTS

- SecureFlow: Designed an intraprocedural, field-sensitive taint-analysis engine with a JSON ruleset (sinks/sanitizers/severity); validated on a benchmark suite, correctly flagging command-injection while suppressing false positives on sanitized input
- NodeSync: Engineered a fault-tolerant replicated key-value store in Python (3-node TCP cluster) with heartbeat failure detection, Raft-lite leader election, and runtime-switchable consistency modes; benchmarked eventual (0.0139s) vs. strong (0.0153s) consistency with zero-downtime failover
- PyScope: Developed a script-level profiler with wall-clock/CPU/memory sampling and function-level hotspot detection, plus a historical regression detector flagging execution-time, memory, and hotspot regressions across runs
- Py2C: Implemented a compiler pipeline (AST $\rightarrow$ custom IR $\rightarrow$ optimizer $\rightarrow$ C codegen) with constant-folding and dead-code-elimination passes, supporting functions, full control flow, and printf lowering
- ModelTrace: Architected a PyTorch inspection framework using forward hooks to compute layer-wise sparsity and activation drift, plus a composite robustness score attributing output shifts to individual layers
- CodeChecker: Built a Java linter combining style checks with method-level cyclomatic-complexity analysis and CI-integrated JSON output; a self-built complement to 100+ PRs merged into Checkstyle

SKILLS

Languages: Java, Python, Solidity, C++, JavaScript

Systems & Tools: Linux, Git, Docker, Maven, CI/CD, Flask, React.js, Scikit-learn, Pandas, Web3.js

Core Areas: Software Security, Static Analysis, Distributed Systems, Applied ML, NLP, Smart Contracts & Governance

HONORS AND AWARDS

- JN Tata Scholar: J.N. Tata Endowment (Tata Trusts), India's oldest merit-based scholarship for postgraduate study abroad (est. 1892, ~5,800 scholars to date); selected via a competitive, interview-based national process
- See Yourself in Cyber Scholarship: NCA & ISACA Foundation; \$2,000 award, selected as 1 of 12 recipients (July 2026)
- ISSA Education Foundation General Scholarship: sponsored by Docent Institute; \$2,000 award (August 2026)
- IELTS Academic: Overall Band 8.0 (CEFR C1), January 2026